

EXPERIMENT 11 :

ANCOVA (Analysis of Covariance)

OBJECTIVE:

Write a Program on conducting Analysis of Covariance (ANCOVA) with categorical variables on the iris dataset involves multiple steps.

THEORY:

Analysis of Covariance (ANCOVA) is a statistical technique that combines ANOVA and regression to compare group means while controlling for the effect of one or more continuous covariates. It adjusts the dependent variable for the influence of covariates, providing a more accurate comparison between categorical groups.

CODE:

```
# =====
# ANCOVA Analysis on Iris Dataset (Error-Free Version)
# =====

# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf

# -----
# Load dataset from sklearn (more reliable than seaborn)
# -----
from sklearn.datasets import load_iris

iris_data = load_iris()
iris = pd.DataFrame(iris_data.data, columns=iris_data.feature_names)

# Rename columns for simplicity
iris.columns = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width']

# Add species column
iris['species'] = iris_data.target
iris['species'] = iris['species'].map({0: 'setosa', 1: 'versicolor', 2:
'virginica'})

print("Dataset Loaded Successfully!\n")
print(iris.head())

# -----
# Convert species to categorical
# -----
iris['species'] = iris['species'].astype('category')

# -----
# Visualization
# -----
sns.lmplot(x='sepal_width', y='sepal_length', hue='species', data=iris)
plt.title("Sepal Length vs Sepal Width by Species")
plt.show()

# -----
# ANCOVA Model
# -----
model = smf.ols('sepal_length ~ species + sepal_width', data=iris).fit()
```

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# -----
# ANCOVA Table
# -----
ancova_table = sm.stats.anova_lm(model, typ=2)

print("\n===== ANCOVA TABLE =====\n")
print(ancova_table)

# -----
# Model Summary
# -----
print("\n===== MODEL SUMMARY =====\n")
print(model.summary())

# -----
# Interaction Effect (Optional)
# -----
interaction_model = smf.ols('sepal_length ~ species * sepal_width',
data=iris).fit()
interaction_table = sm.stats.anova_lm(interaction_model, typ=2)

print("\n===== ANCOVA WITH INTERACTION =====\n")
print(interaction_table)

# =====
# END
# =====
# -----
# ANCOVA Analysis on Iris Dataset (Error-Free Version)
# =====

# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf

# -----
# Load dataset from sklearn (more reliable than seaborn)
# -----
from sklearn.datasets import load_iris

iris_data = load_iris()
iris = pd.DataFrame(iris_data.data, columns=iris_data.feature_names)

# Rename columns for simplicity
iris.columns = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width']

# Add species column
iris['species'] = iris_data.target
iris['species'] = iris['species'].map({0: 'setosa', 1: 'versicolor', 2:
'virginica'})

print("Dataset Loaded Successfully!\n")
print(iris.head())

# -----
# Convert species to categorical
# -----
iris['species'] = iris['species'].astype('category')

# -----
# Visualization
# -----
sns.lmplot(x='sepal_width', y='sepal_length', hue='species', data=iris)

```

```

plt.title("Sepal Length vs Sepal Width by Species")
plt.show()

# -----
# ANCOVA Model
# -----
model = smf.ols('sepal_length ~ species + sepal_width', data=iris).fit()

# -----
# ANCOVA Table
# -----
ancova_table = sm.stats.anova_lm(model, typ=2)

print("\n===== ANCOVA TABLE =====\n")
print(ancova_table)

# -----
# Model Summary
# -----
print("\n===== MODEL SUMMARY =====\n")
print(model.summary())

# -----
# Interaction Effect (Optional)
# -----
interaction_model = smf.ols('sepal_length ~ species * sepal_width',
data=iris).fit()
interaction_table = sm.stats.anova_lm(interaction_model, typ=2)

print("\n===== ANCOVA WITH INTERACTION =====\n")
print(interaction_table)

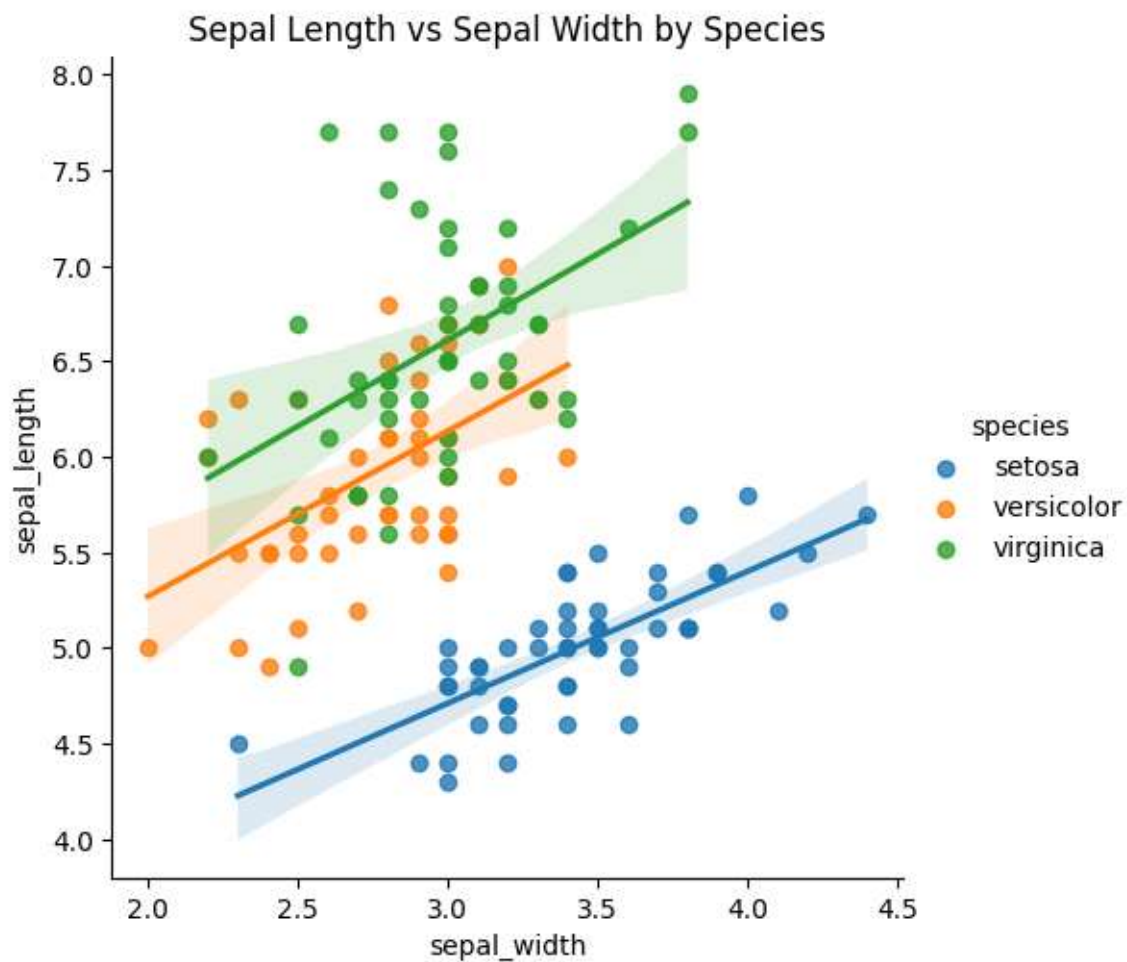
# =====
# END
# =====

```

OUTPUT:

Dataset Loaded Successfully!

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa



===== ANCOVA TABLE =====

	sum_sq	df	F	PR(>F)
species	72.752431	2.0	189.651157	2.557870e-41
sepal_width	10.952535	1.0	57.102173	4.187340e-12
Residual	28.003665	146.0	NaN	NaN

===== MODEL SUMMARY =====

OLS Regression Results

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```

Dep. Variable:          sepal_length    R-squared:
0.726
Model:                  OLS             Adj. R-squared:
0.720
Method:                 Least Squares    F-statistic:
128.9
Date:                   Fri, 27 Mar 2026  Prob (F-statistic):
7.66e-41
Time:                   07:04:46         Log-Likelihood:
86.968
No. Observations:      150              AIC:
181.9
Df Residuals:          146              BIC:
194.0
Df Model:               3
Covariance Type:       nonrobust

```

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=====
                                coef      std err          t      P>|t|
[0.025      0.975]
-----
Intercept                    2.2514      0.370      6.089      0.000
1.521      2.982
species[T.versicolor]      1.4587      0.112     13.012      0.000
1.237      1.680
species[T.virginica]      1.9468      0.100     19.465      0.000
1.749      2.144
sepal_width                  0.8036      0.106      7.557      0.000
0.593      1.014
=====
===
Omnibus:                    7.510    Durbin-Watson:
2.066
Prob(Omnibus):              0.023    Jarque-Bera (JB):
7.629
Skew:                       0.423    Prob(JB):
0.0221
Kurtosis:                   3.710    Cond. No.
36.4
=====
===

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

===== ANCOVA WITH INTERACTION =====

	sum_sq	df	F	PR(>F)
species	72.752431	2.0	188.109117	6.866547e-41
sepal_width	10.952535	1.0	56.637879	5.221114e-12
species:sepal_width	0.157194	2.0	0.406442	6.667773e-01
Residual	27.846471	144.0	NaN	NaN

